US 20250286572A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0286572 A1
(43) Pub. Date: Sep. 11, 2025
Srirattana et al.

(54) COUPLER BYPASS ARCHITECTURE FOR PASS-THROUGH INSERTION LOSS IMPROVEMENT

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(21) Appl. No.: **19/071,837**

(22) Filed: **Mar. 6, 2025**

Related U.S. Application Data

(60) Provisional application No. 63/561,818, filed on Mar. 6, 2024.

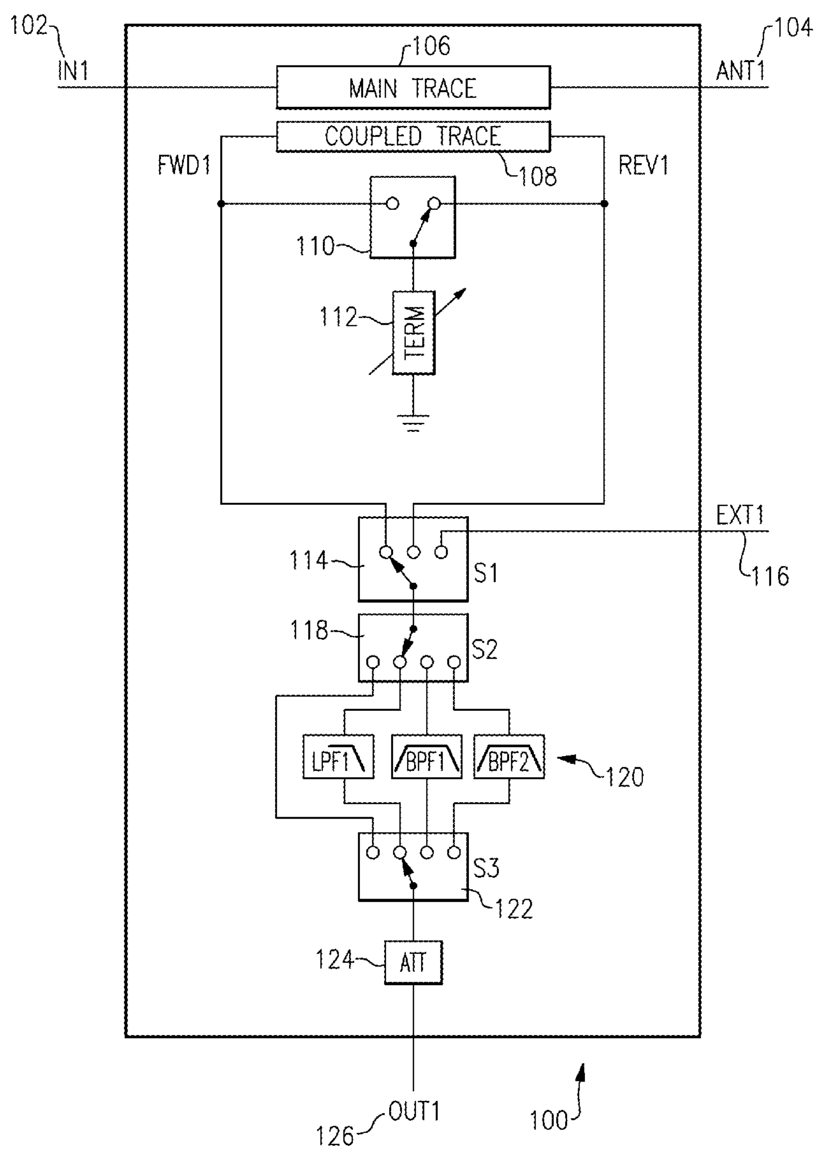
Publication Classification

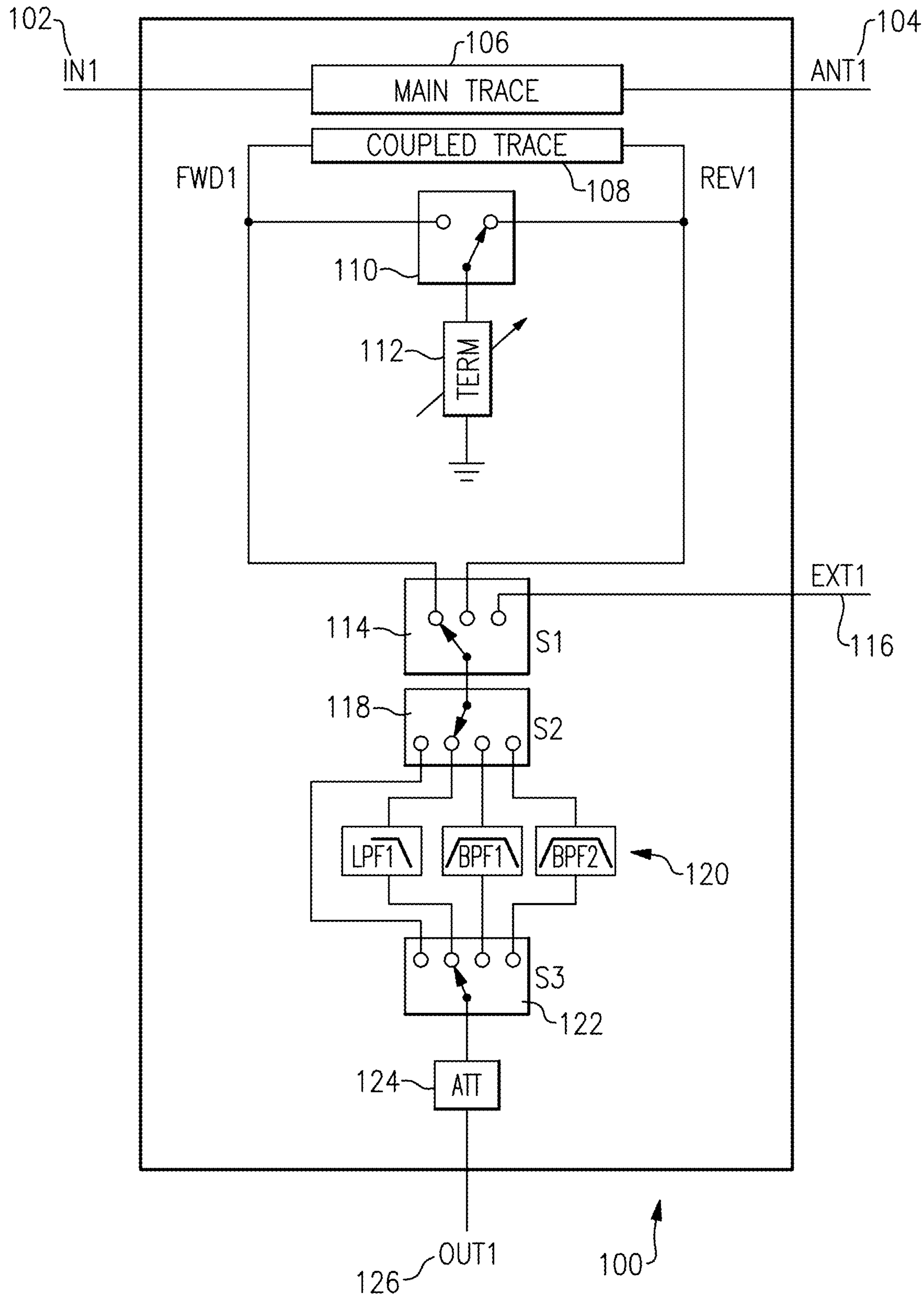
(51) Int. Cl.
H04B 1/16 (2006.01)

(52) U.S. Cl.
CPC *H04B 1/1638* (2013.01)

(57) **ABSTRACT**

A module is presented for an electronic telecommunications device, the module including: a forward input; a reverse input; a bypass input; an output; a first plurality of switches coupled between the forward input, the reverse input, the bypass input, and the output, the first plurality of switches configured to selectively couple at least one of the forward input, reverse input, or bypass input to the output; and at least one bypass switch configured to selectively couple the bypass input to the output.

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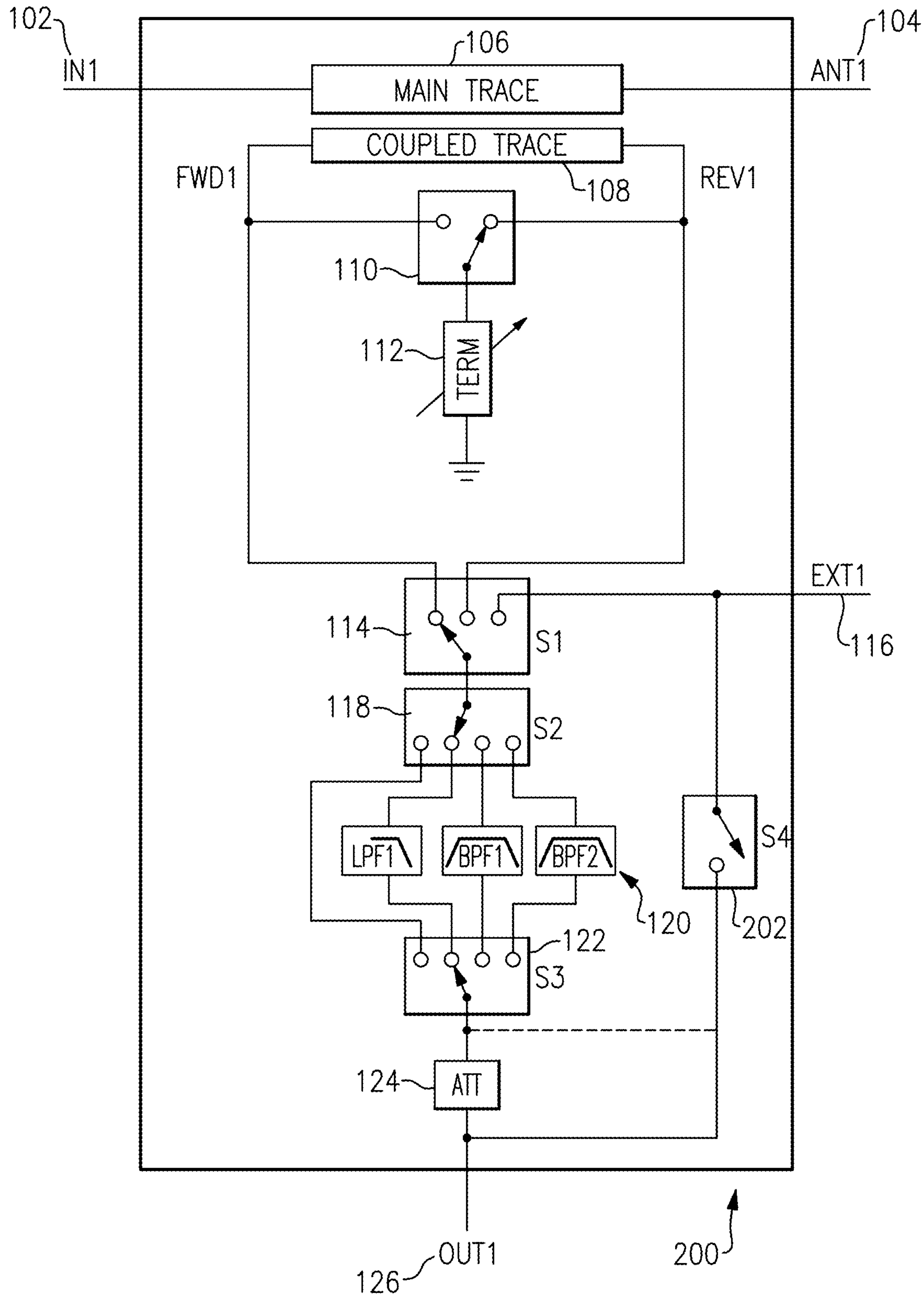
IN1
106
MAIN TRACE
104
ANT1
COUPLED TRACE
FWD1
108
REV1
110
112
TERM
EXT1
114
S1
116
118
S2
LPF1
BPF1
BPF2
120
S3
122
124
ATT
126
OUT1
100

FIG.1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108
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TERM
EXT1
114
S1
116
118
S2
LPF1
BPF1
BPF2
S4
120
202
122
S3
124
ATT
200
126
OUT1

FIG.2

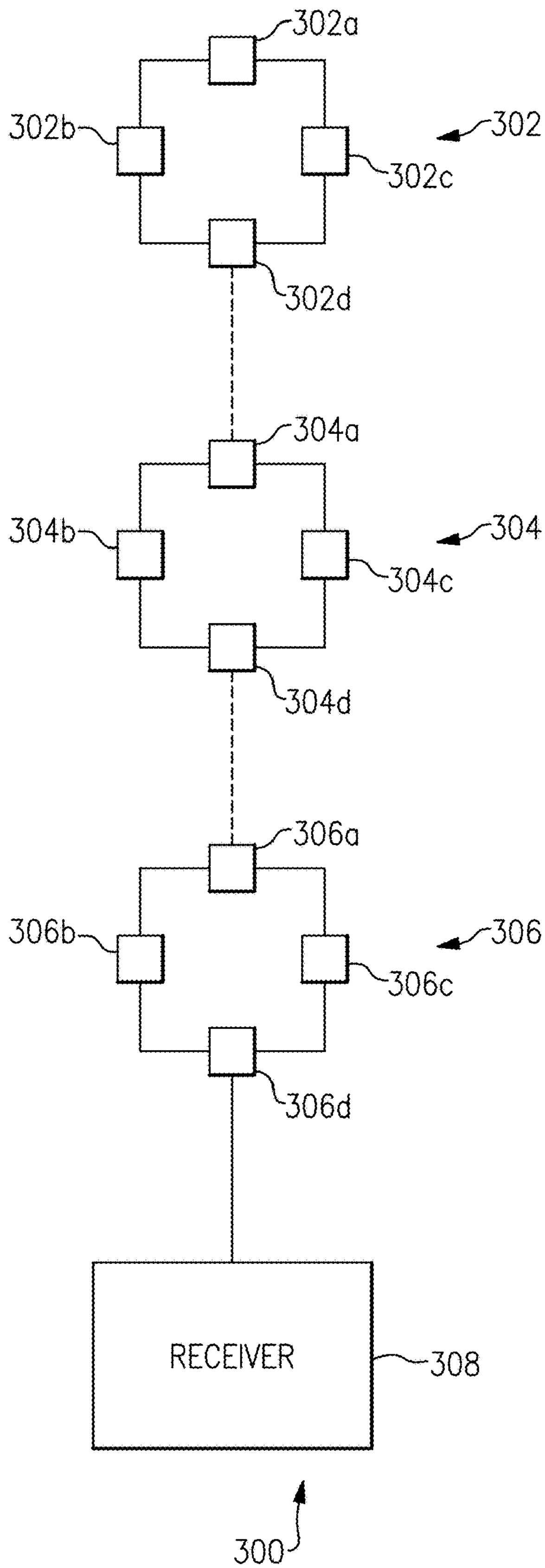
302a
302b
302
302c
302d
304a
304b
304
304c
304d
306a
306b
306
306c
306d
RECEIVER
308
300

FIG.3

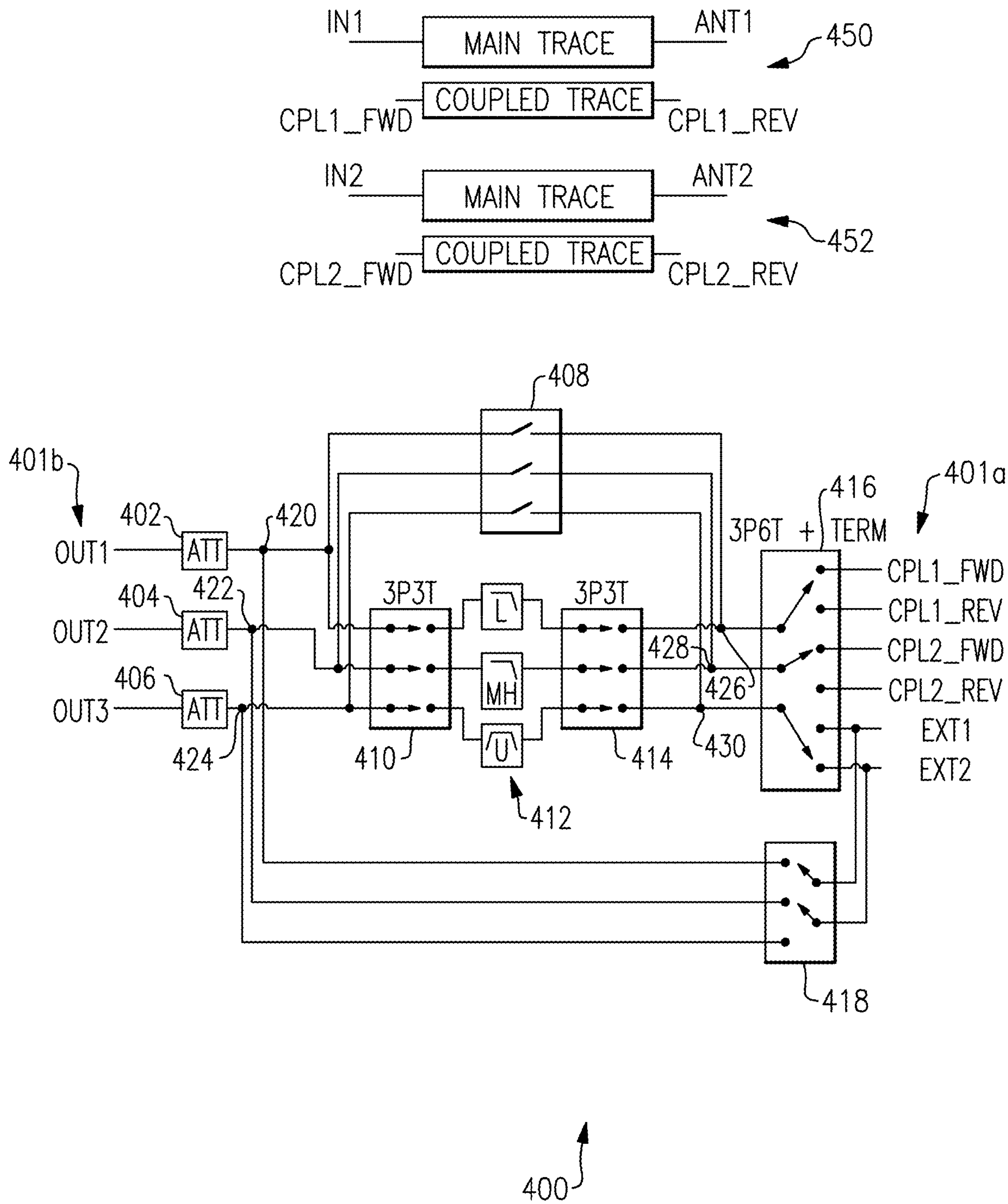
IN1
MAIN TRACE
ANT1
450
COUPLED TRACE
CPL1_FWD
CPL1_REV
IN2
MAIN TRACE
ANT2
452
COUPLED TRACE
CPL2_FWD
CPL2_REV
408
401b
401a
416
3P6T + TERM
402
ATT
420
OUT1
404
422
ATT
OUT2
3P3T
3P3T
L
MH
U
428
426
406
ATT
OUT3
424
410
414
430
412
CPL1_FWD
CPL1_REV
CPL2_FWD
CPL2_REV
EXT1
EXT2
418
400

FIG.4

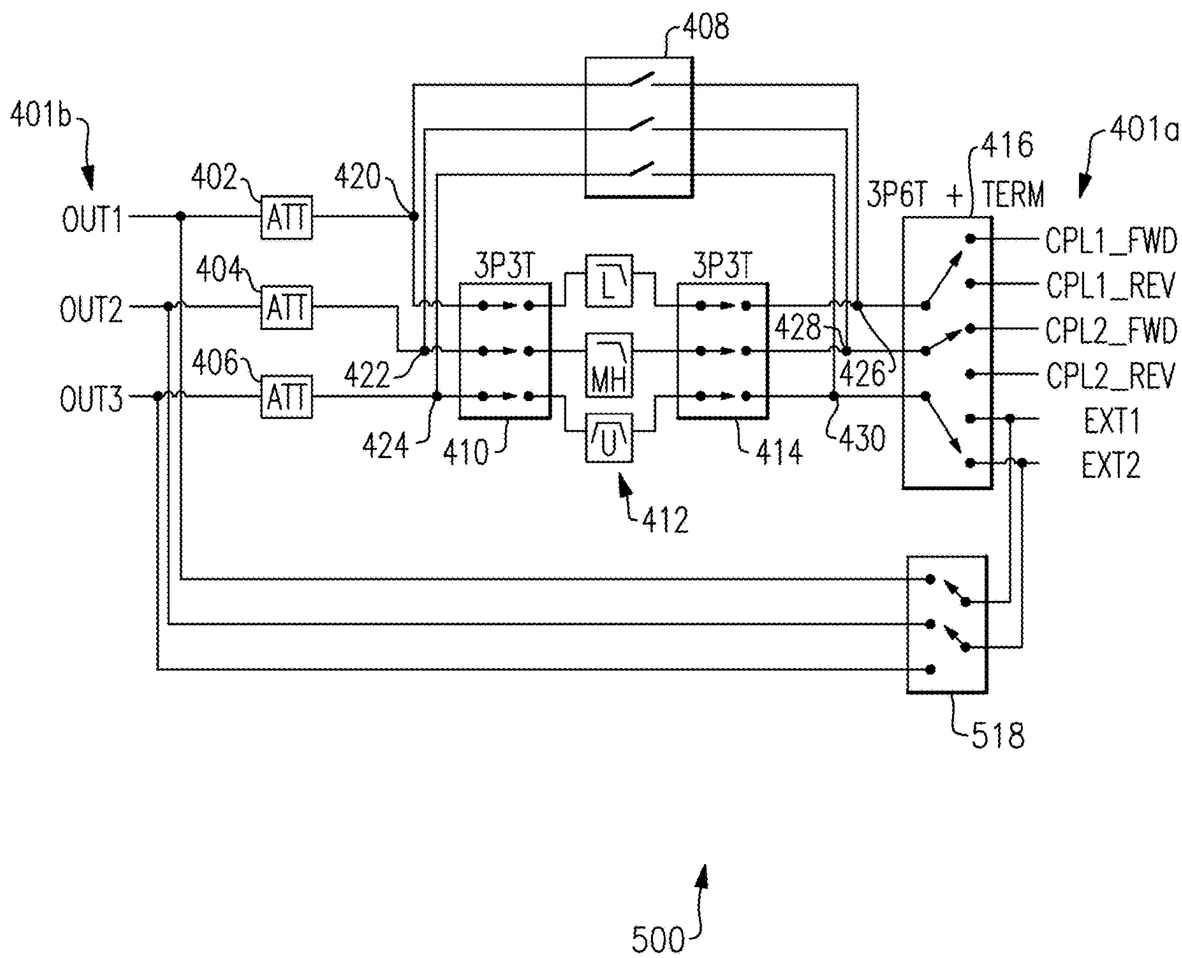
408
401b
401a
416
3P6T + TERM
OUT1
OUT2
OUT3
402
404
406
ATT
ATT
ATT
420
422
424
3P3T
3P3T
410
414
L
MH
U
412
428
426
430
CPL1_FWD
CPL1_REV
CPL2_FWD
CPL2_REV
EXT1
EXT2
518
500

FIG.5

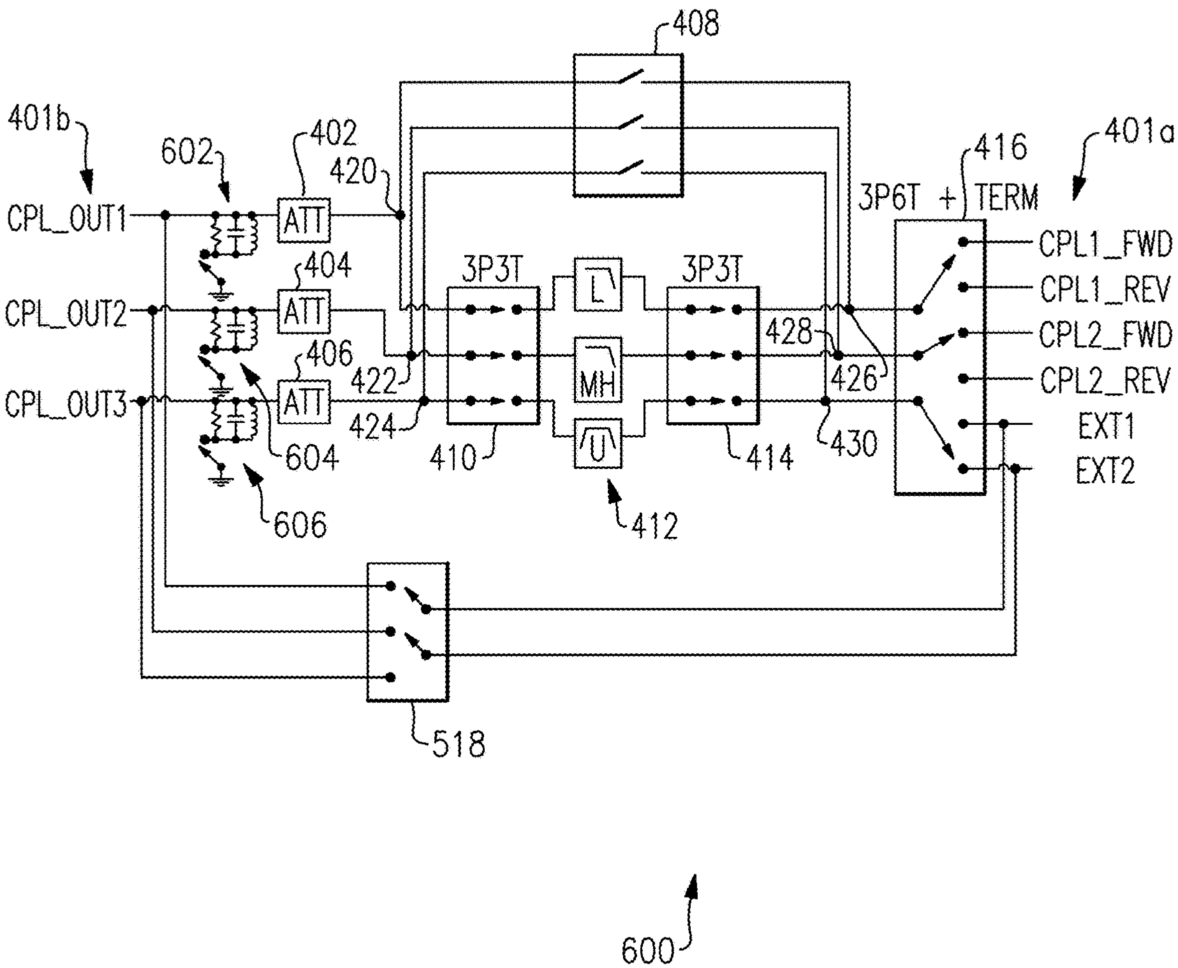
408
401b
602
402
420
416
401a
3P6T + TERM
CPL_OUT1
ATT
CPL1_FWD
404
3P3T
3P3T
CPL1_REV
CPL_OUT2
ATT
428
CPL2_FWD
406
L
MH
422
426
CPL2_REV
CPL_OUT3
ATT
424
U
430
EXT1
604
410
414
EXT2
606
412
518
600

FIG.6

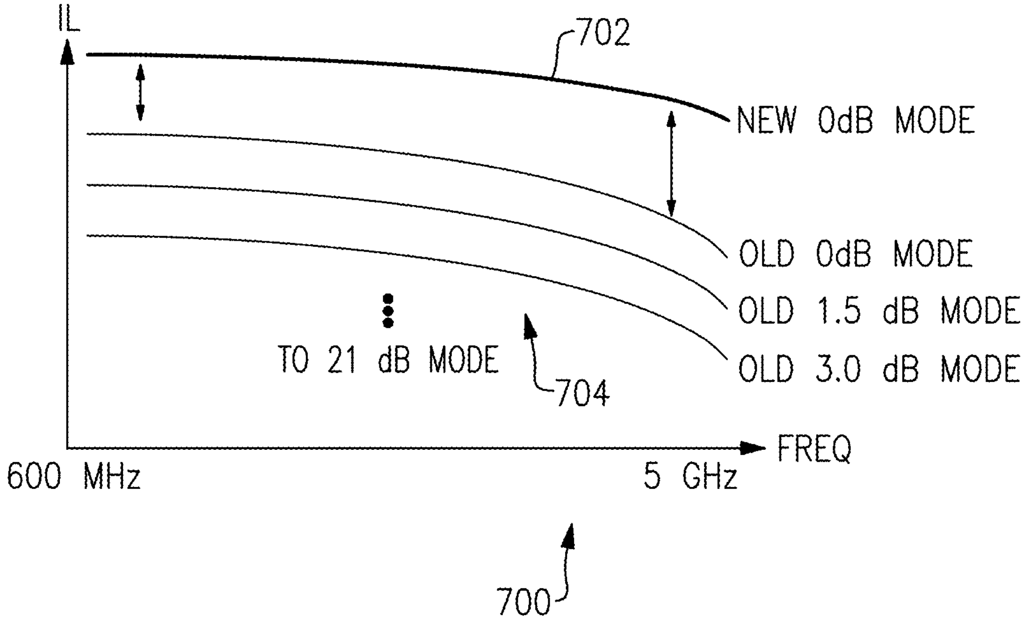
IL
702
NEW 0dB MODE
OLD 0dB MODE
OLD 1.5 dB MODE
OLD 3.0 dB MODE
TO 21 dB MODE
704
FREQ
600 MHz
5 GHz
700

FIG.7A

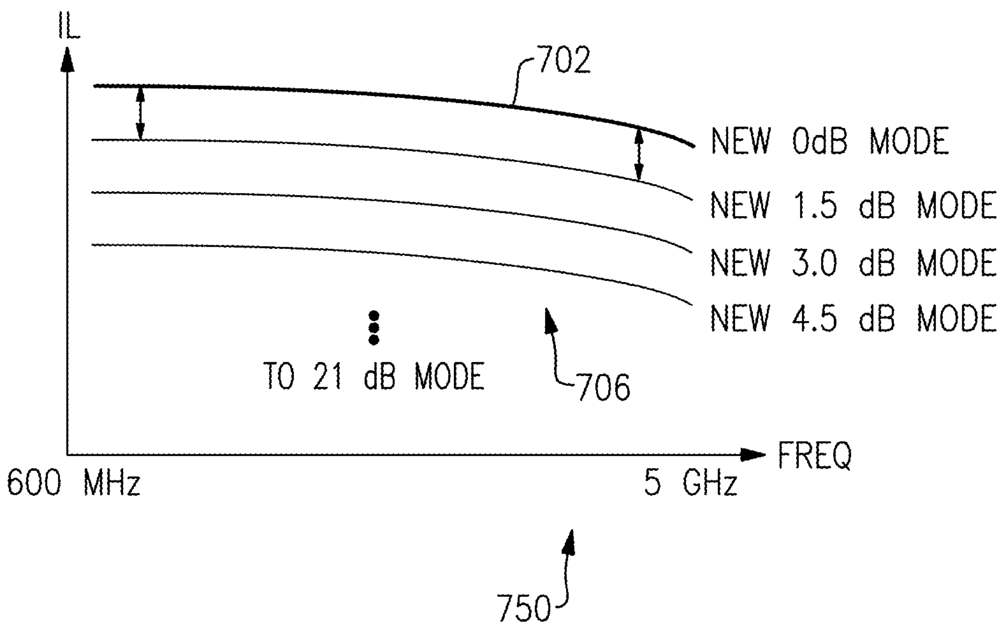
IL
702
NEW 0dB MODE
NEW 1.5 dB MODE
NEW 3.0 dB MODE
NEW 4.5 dB MODE
TO 21 dB MODE
706
FREQ
600 MHz
5 GHz
750

FIG.7B

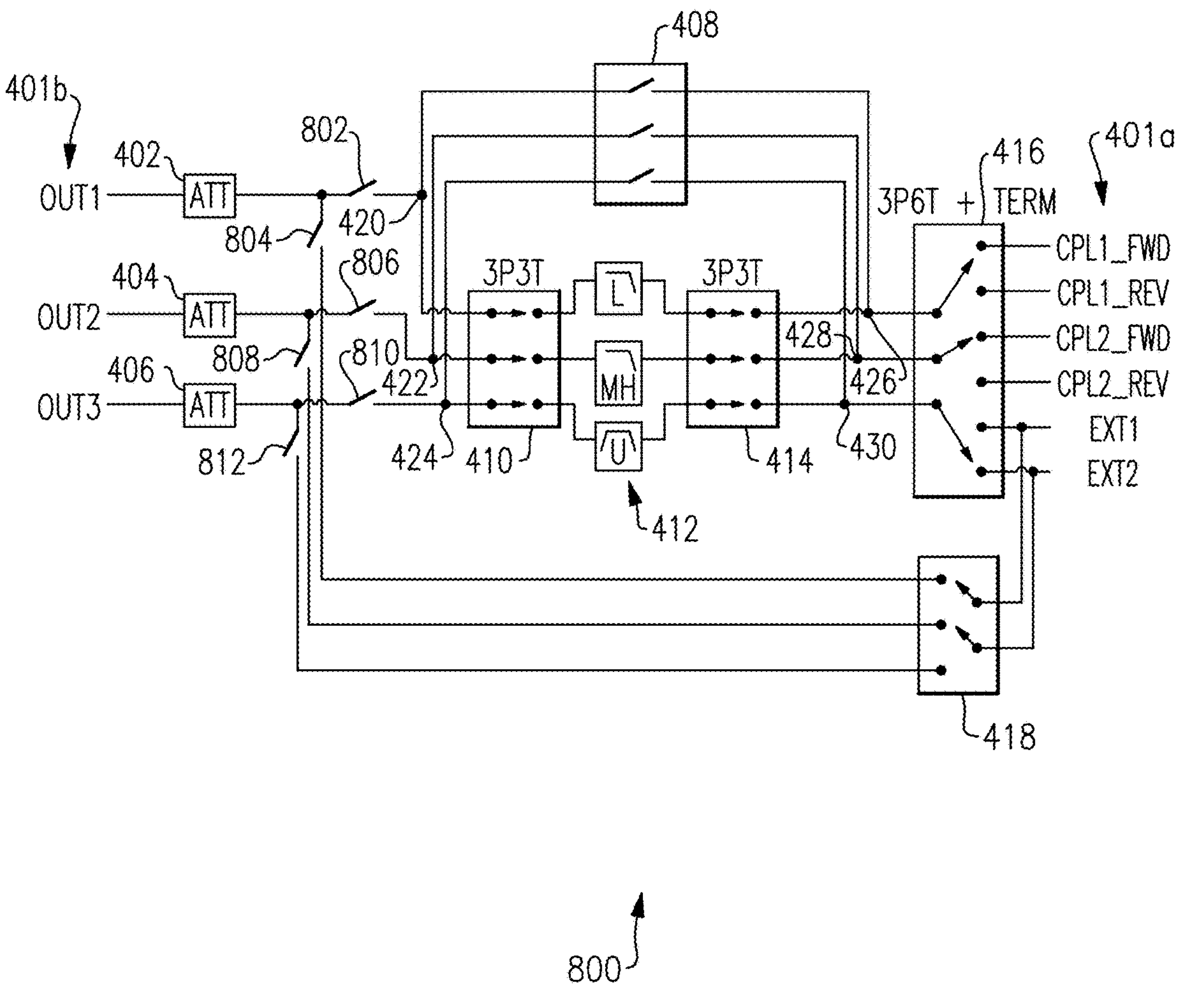
408
401b
402
802
OUT1
ATT
804
420
416
401a
3P6T + TERM
404
806
3P3T
3P3T
CPL1_FWD
CPL1_REV
OUT2
ATT
L
808
810
MH
428
CPL2_FWD
406
422
426
CPL2_REV
OUT3
ATT
812
424
410
U
414
430
EXT1
EXT2
412
418
800

FIG.8

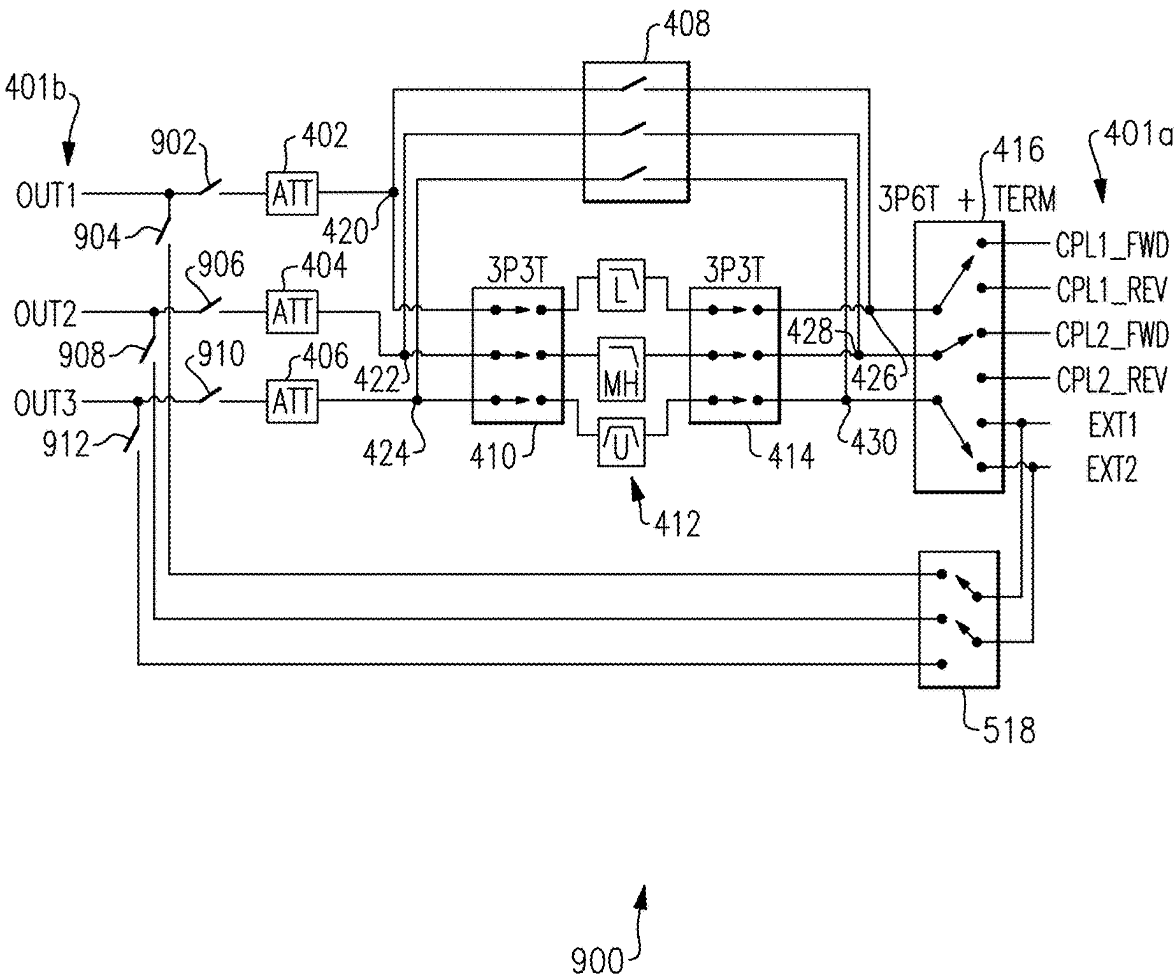
408
401b
902
402
416
401a
3P6T + TERM
OUT1
ATT
904
420
CPL1_FWD
906
404
3P3T
3P3T
CPL1_REV
OUT2
ATT
L
CPL2_FWD
908
428
910
406
422
MH
426
CPL2_REV
OUT3
ATT
EXT1
912
424
410
U
414
430
EXT2
412
518
900

FIG.9

COUPLER BYPASS ARCHITECTURE FOR PASS-THROUGH INSERTION LOSS IMPROVEMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/561,818, titled COUPLER BYPASS ARCHITECTURE FOR PASS-THROUGH INSERTION LOSS IMPROVEMENT, filed on Mar. 6, 2024, which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

[0002] At least one example in accordance with the present disclosure relates generally reducing insertion loss in coupled systems.

SUMMARY

[0003] According to at least one aspect of the present disclosure a module for an electronic telecommunications device is presented, the module comprising: a forward input; a reverse input; a bypass input; an output; a first plurality of switches coupled between the forward input, the reverse input, the bypass input, and the output, the first plurality of switches configured to selectively couple at least one of the forward input, reverse input, or bypass input to the output; and at least one bypass switch configured to selectively couple the bypass input to the output.

[0004] In some examples, the first plurality of switches includes a first switch, a second switch, and a third switch, the first switch being coupled to the forward input, reverse input, and bypass input, the second switch being coupled to the first switch and the third switch, and the third switch being coupled between the second switch and the output. In some examples, the module further comprises a plurality of filters coupled to the second switch and the third switch, wherein the second switch is configured to selectively couple the first switch to a filter of the plurality of filters or to the third switch, and the third switch is configured to selectively couple the filter of the plurality of filters or the second switch to the output. In some examples, the module further comprises an attenuator coupled between the plurality of switches and the output, wherein the at least one bypass switch is coupled either between the plurality of switches and the attenuator, or between the attenuator and the output. In some examples, the module further comprises one or more additional outputs and one or more additional attenuators, the one or more additional attenuators being coupled between the plurality of switches and the one or more additional outputs; and one or more additional bypass inputs coupled to the at least one bypass switch, the at least one bypass switch being configured to selectively couple at least one of the one or more additional bypass inputs to the output. In some examples, the at least one bypass switch is further coupled either between the one or more additional attenuators and the plurality of switches or between the one or more additional attenuators and the one or more additional outputs. In some examples, the module further comprises a plurality of impedance networks, each impedance network coupled to a respective attenuator or additional attenuator and to a respective output or additional output. In some examples, the module further comprises an impedance network coupled between the attenuator and the output. In some examples, the module further comprises a second plurality of switches coupled to the output, the second plurality of switches including at least one switch coupled between the output and the at least one bypass switch and at least one switch coupled between the output and the plurality of switches. In some examples, the module further comprises an attenuator coupled either between the second plurality of switches and the output or between the second plurality of switches and the plurality of switches. In some examples, the plurality of switches includes a first switch, a second switch, a third switch, and a fourth switch, the first switch coupled to the second switch, the second switch coupled to the third switch, the third switch coupled to the output and the first switch coupled to at least one of the forward input, reverse input, or bypass input, and the fourth switch coupled to the first switch and to the output. In some examples, the module further comprises a first trace configured to receive a signal; a second trace electromagnetically coupled to the first trace and configured to provide an induced signal responsive to the first trace receiving the signal, wherein the second trace is coupled at a first end to the forward input and at a second end to the reverse input.

[0005] According to at least one aspect of the present disclosure a system of connected modules is presented, the system comprising: a first module including a first forward input, a first reverse input, a first bypass input, a first output, a first plurality of switches coupled between the first forward input, first reverse input, and first output, and at least one first bypass switch coupled between the first bypass input and the first output; and a second module including a second forward input, a second reverse input, a second bypass input coupled to the first output, a second output, a second plurality of switches coupled between the second forward input, second reverse input, and second output, and at least one second bypass switch coupled between the second bypass input and the second output.

[0006] According to at least one aspect of the present disclosure a system of connected modules is presented, the system comprising: a plurality of modules coupled together in a series, each respective module of the plurality of modules having a forward input, a reverse input, a bypass input, an output, a plurality of switches coupled between the forward input, the reverse input, and the output of the respective module, and at least one bypass switch coupled between the bypass input and the output of the respective module, the output of a first module of the plurality of modules in the series being coupled to the bypass input of a second module of the plurality of modules in the series.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Various aspects of at least one embodiment are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide an illustration and a further understanding of the various aspects and embodiments, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the limits of any particular embodiment. The drawings, together with the remainder of the specification, serve to explain principles and operations of the described and claimed aspects and embodiments. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every figure. In the figures:

[0008] FIG. **1** illustrates a signal processing module according to an example;

[0009] FIG. **2** illustrates a signal processing module according to an example;

[0010] FIG. **3** illustrates a plurality of signal processing modules according to an example;

[0011] FIG. **4** illustrates a signal processing module according to an example;

[0012] FIG. **5** illustrates a signal processing module according to an example;

[0013] FIG. **6** illustrates a signal processing module according to an example;

[0014] FIG. **7**A illustrates a graph of transfer functions of a signal processing module;

[0015] FIG. **7**B illustrates a graph of transfer functions of a signal processing module;

[0016] FIG. **8** illustrates a signal processing module according to an example; and

[0017] FIG. **9** illustrates a signal processing module according to an example.

DETAILED DESCRIPTION

[0018] In various electronic applications, including telecommunications, coupled signals may be provided to a given circuit or circuit element (called a "receiver"), such as a feedback receiver, after passing through multiple modules configured to process a signal. Furthermore, each module may be associated with a different antenna (and therefore, a different coupled signal). Thus, a first module may be configured to process a 5 GHz signal, a second module may be configured to process a 6 GHz signal, and so forth.

[0019] However, when the modules are coupled together, a given coupled signal may have to pass through multiple modules to reach the receiver. Each module may, therefore, apply some processing to a given signal. For example, a signal originating in a module configured to process a 5 GHz signal may pass through a module configured to process a 6 GHz signal and may undergo adjustments intended for the 6 GHz signal but not intended for the 5 GHz signal. Therefore, modules may include bypass switches that allow a signal to bypass the processing elements (e.g., filters, amplifiers, and so forth) of a given module. By bypassing the processing elements of the module, the signal can be provided to the receiver relatively unchanged compared to if the signal passes through the processing elements of the module.

[0020] However, each switch introduces insertion loss associated with the switch. Consider the module **100** of FIG. **1**. A coupled signal on the coupled trace **108** must, to reach the output **126**, pass through the input selection switch **114**, the filter selection switch **118**, and the output selection switch **122**. A signal originating at the second input **116** must pass through the same three switches. Each of these switches (the input selection switch **114**, the filter selection switch **118**, and the output selection switch **122**) will introduce some insertion loss.

[0021] Now, suppose multiple modules similar to the module **100** of FIG. **1** are connected in a "daisy-chain," such that the output (corresponding to output **126**) of a first module is connected to the second input (corresponding to second input **116**) of a second module, the output of the second module is coupled to the second input of a third module, and so forth, for a total of n modules, where n is any integer. Each module has an input selection switch **114**, a filter selection switch **118**, and an output selection switch **122**. Thus, a signal originating with the first module in the chain must go through n sets of three switches, with each switch causing some insertion loss, before reaching the receiver circuit coupled to the end of the chain. Additionally, the signal may be further attenuated by the attenuation element **124** of each module in the chain (though the attenuation element **124** may be in zero dB gain mode and thus may not necessarily attenuate the signal).

[0022] Aspects and elements of the present disclosure relate to techniques and systems for reducing the attenuation of a signal as it is provided to intercoupled modules (such as module **100** of FIG. **1**). In some examples, the attenuation that is reduced is the attenuation caused by the switches within the modules (e.g., the insertion loss is reduced). In some examples, the insertion loss is reduced by reducing the number of switches along the path between the signal's origin and the receiver.

[0023] Returning to the discussion of FIG. **1**, FIG. **1** illustrates a module **100** for use in telecommunications according to an example. A signal received at the input **102** and/or antenna connection **104** passes through the main trace **106**, inducing a signal on the coupled trace **108**, which is routed via the switches **114**, **118**, **122** to the output **126**.

[0024] The module **100** includes a first input **102**, an antenna connection **104**, a main trace **106**, a coupled trace **108**, a mode selection switch **110**, a termination impedance **112**, a source selection switch **114**, a filter selection switch **118**, a second input **116**, a plurality of filtering element **120** ("filters **120**"), an output selection switch **122**, an attenuation element **124**, and an output **126**.

[0025] The main trace **106** is coupled to the first input **102** and the antenna connection **104**. The main trace **106** is further electromagnetically coupled to the coupled trace **108**. The coupled trace **108** is coupled to the mode selection switch **110** and the source selection switch **114**. The mode selection switch **110** is coupled to the termination impedance **112**, and the termination impedance **112** may be coupled to a reference node. The source selection switch **114** is coupled to the filter selection switch **118** and to the second input **116**. The filter selection switch **118** is coupled to the filters **120** and to the output selection switch **122**. The filters **120** are coupled to the output selection switch **122**. The output selection switch **122** is coupled to the attenuation element **124**, and the attenuation element **124** is coupled to the output **126**.

[0026] The first input **102** and antenna connection **104** may be configured to receive a communication signal, such as a sinusoidal voltage and/or current produced by an electrical or magnetic field and/or changes in an electrical or magnetic field. The communication signal may be received via a wired or wireless connection. The communication signal may pass from the antenna connection **104** to the first input **102** or from the first input **102** to the antenna connection **104**. In either case, the communication signal may pass through the main trace **106**.

[0027] The main trace **106** is electromagnetically coupled to the coupled trace **108**. When a current and/or voltage is present on the main trace **106**, a corresponding voltage and/or current may be induced on the coupled trace **108**. The coupled voltage and/or current ("coupled signal") may be proportionate or otherwise based on the current and/or voltage present on the main trace **106** and/or may be related to changes in the current and/or voltage present on the main trace.

[0028] The module **100** may have a forward or reverse mode of operation in some examples. This means that the communication signal that induces the coupled signal may be expressed as a current flowing from the first input **102** to the antenna connection **104**, or from the antenna connection **104** to the first input **102**. The state of the mode selection switch **110** may change based on the mode of operation. For example, a first connection of the mode selection switch **110** may be coupled to a first end (FWD**1**) of the coupled trace **108**, and a second connection of the mode selection switch **110** may be coupled to a second end (REV**1**) of the coupled trace **108**. The mode selection switch **110** may be configured to selectively couple either the first connection (and therefore the first end of the coupled trace **108**) or the second connection (and therefore the second end of the coupled trace **108**) to the termination impedance **112**. For example, in the forward mode of operation, the mode selection switch **110** may be configured to selectively couple the second connection to the termination impedance **112**, and in the reverse mode of operation, the mode selection switch **110** may be configured to selectively couple the first connection to the termination impedance **112**. The termination impedance **112** may be variable or constant, and may include resistive and/or reactive elements.

[0029] The coupled trace **108** may also be coupled to the source selection switch **114**. In some examples, the first end (FWD**1**) of the coupled trace **108** is coupled to a first connection of the source selection switch **114**, the second end (REV**1**) of the coupled trace **108** is coupled to a second connection of the source selection switch **114**, and the second input **116** is coupled to a third connection of the source selection switch **114**. The source selection switch **114** is, in turn, configured to selectively couple one of the first connection, second connection, or third connection of the source selection switch **114** to the filter selection switch **118**.

[0030] The filter selection switch **118** in turn has a plurality of connections, including at least a first connection and a second connection. The filter selection switch **118** is configured to selectively couple the input selection switch **114** to either the output selection switch **122** or a filter of the filters **120**. The first connection of the filter selection switch **118** is coupled to a corresponding connection of the output selection switch **122**. The second connection of the filter selection switch **118** is coupled to a first filter of the filters **120**. Each additional connection of the filter selection switch **122** is coupled to a corresponding filter of the filters **120**. For example, a third connection of the filter selection switch **118** may be coupled to a second filter of the filters **120**, and a fourth connection of the filter selection switch **118** may be coupled to a third filter of the filters **120**.

[0031] The filters **120** may include one or more filters configured to condition, attenuate, and/or process the coupled signal. For example, the filters **120** may include one or more low-pass, high-pass, band-pass, band-reject, and/or other filters. In general, the filter selection switch **118** and the output selection switch **122** have at least one connection each corresponding to each respective filter of the filters **120**.

[0032] The output selection switch **122** is configured to selectively couple the filter selection switch **118** or one filter of the filters **120** to the attenuation element **124**. The attenuation element **124** is configured to process (e.g., provide attenuation, gain, or other changes to) the coupled signal. In some examples, the attenuation element **124** may have a 0 dB mode. That is, the attenuation element **124** may be configured to provide no and/or minimal attenuation or gain.

[0033] The second input **116** is configured to provide a signal to the input selection switch **114**. In some examples, the second input **116** is configured to provide a second coupled signal from (i.e., originating at) a different module.

[0034] Thus, in one example of operation, a communication signal on the main trace **106** induces the coupled signal on the coupled trace **108**. The input selection switch **114** then selects between the coupled signal or the second coupled signal. Whichever of the two possible coupled signals is selectively coupled to the filter selection switch **118** by the input selection switch **114** is then provided to the filter selection switch **118**. The filter selection switch **118** then provides the selected signal to either one of the filters **120** or to the output selection switch **122**. The output selection switch **122** may then provide the selected signal to the attenuation element **124** and thence to the output **126**.

[0035] With reference to the earlier discussion of insertion loss, note that when the second input **116** is selectively coupled (via the input selection switch **114**) to the filter selection switch **118**, and the filter selection switch **118** is selectively coupling the filter selection switch **118** and the attenuation element **124**, the second coupled signal will pass through each of those three switches **114**, **118**, **122**, and may experience insertion loss at each of those switches **114**, **118**, **122**.

[0036] FIG. **2** illustrates a module **200** according to an example. The module **200** is generally similar to the module **100** of FIG. **1**, with some differences. A bypass switch **202** has been added. The bypass switch **202** is coupled to the second input **116** and may be coupled directly to the output **126** or may be coupled to the output **126** via the attenuation element **124**.

[0037] The bypass switch **202** is configured to selectively couple the second input **116** to the output **126** and/or to the attenuation clement **124**. In some examples, when the bypass switch **202** is in a closed state (e.g., coupling the second input **116** to the output **126** and/or attenuation element **124**), at least one of the input selection switch **114**, filter selection switch **118**, and/or output selection switch **122** may be in an open state (e.g., decoupled such that no conducting path is available via the respective switch).

[0038] When a signal, such as the second coupled signal, is provided via the second input **116**, the bypass switch **202** may selectively couple the second input **116** to the attenuation element **124** and/or the output **126**, thereby allowing the signal to avoid insertion loss from the input selection switch **114**, the filter selection switch **118**, and/or the output selection switch **122**.

[0039] FIG. **3** illustrates a system **300** including a plurality of modules and a receiver according to an example. The system **300** includes a first module **302**, a second module **304**, a third module **306**, and a receiver **308**. The system **300** may include one or more additional modules coupled between the second module **304** and the third module **306**. The system **300** may include a receiver **308**.

[0040] The first module **302** includes a first input **302**_a_, a second input **302**_b_, a third input **302**_c_, and an output **302**_d_. The second input **302**_b_ may be an input or output depending on the mode of operation of the module **302** (e.g., forward or reverse modes of operation). The third input **302***c* may be an input or output depending on the mode of operation of the module **302**.

[0041] The second module **304** includes a first input **304***a*, a second input **304***b*, a third input **304***c*, and an output **304***d*. The second input **304***b* may be an input or output depending on the mode of operation of the module **304** (e.g., forward or reverse modes of operation). The third input **304***c* may be an input or output depending on the mode of operation of the module **304**.

[0042] The third module **306** includes a first input **306***a*, a second input **306***b*, a third input **306***c*, and an output **306***d*. The second input **306***b* may be an input or output depending on the mode of operation of the module **306** (e.g., forward or reverse modes of operation). The third input **306***c* may be an input or output depending on the mode of operation of the module **306**.

[0043] Each module **302**, **304**, **306** includes a bypass switch configured to couple the respective first input of that module to a respective output of that respective module. As illustrated, for example, the first input **304***a* of the second module **304** is coupled to the output **302***d* of the first module **302**, while the first input **306***a* of the third module may be coupled to the output **304***d* of the second module **304**. The modules **302**, **304**, **306** may therefore have bypass switches configured to selectively couple their respective first inputs **302***a*, **304***a*, **306***a* to their respective outputs **302***d*, **304***d*, **306***d*. For example, a bypass switch of the second module **304** may directly couple the first input **304***a* of the second module **304** to the output **304***d* of the second module **304**, bypassing some or all of the internal circuit elements and/or electronics of the second module **304**.

[0044] Thus, when a signal originating with the first module **302** is passed to the receiver, the insertion loss experienced by the signal with respect to the second module **304**, third module **306**, and any intervening modules between the second module **304** and third module **306** may be proportionate to the insertion loss of the bypass switches within those respective modules. For example, if there are n modules with identical bypass switches having insertion loss L, the total insertion loss may be based on and/or proportional to nL.

[0045] FIG. **4** illustrates a module topology **400** ("module **400**") according to an example. The module **400** is configured to allow a signal to bypass numerous switches when received at one of the inputs. The module **400** includes a plurality of inputs **401***a*, a plurality of outputs **401***b*, a first attenuator **402**, a second attenuator **404**, a third attenuator **406**, a first plurality of switches **408**, a second plurality of switches **410**, a plurality of filters **412**, a third plurality of switches **414**, a fourth plurality of switches **416**, and a fifth plurality of switches **418**. FIG. **4** also includes a first node **420**, a second node **422**, a third node **424**, a fourth node **426**, a fifth node **428**, and a sixth node **430**. FIG. **4** also illustrates a first pair of traces **450** and a second pair of traces **452**.

[0046] The plurality of outputs **401***b* are coupled to the attenuators **402**, **404**, **406**. In some examples, a first output of the plurality of outputs **401***b* is coupled to the first attenuator **402**, a second output of the plurality of outputs **401***b* is coupled to the second attenuator **404**, and a third output of the plurality of outputs **401***b* is coupled to the third attenuator **406**. The first attenuator **402** is coupled to the first node **420**, the second attenuator **404** is coupled to the second node **422**, and the third attenuator **406** is coupled to the third node **424**.

[0047] Respective switches of the first plurality of switches **408** are coupled to one or more of the first node **420**, second node **422**, third node **424**, fourth node **426**, fifth node **428**, and/or sixth node **430**. In some examples, for the first plurality of switches **408**, a first switch is coupled at a first connection to the first node **420** and at a second connection to the fourth node **426**, a second switch is coupled at a first connection to the second node **422** and at a second connection to the fifth node **428**, and a third switch is coupled to the third node **420** at a first connection and to the sixth node **430** at a second connection.

[0048] Respective switches of the second plurality of switches **410** are coupled to either the first node **420**, second node **422**, third node **424**, and/or at least one respective filter of the plurality of filters **412**. In some examples, for the second plurality of switches **410**, a first switch is coupled at a first connection to the first node **420** and at a second connection to a respective filter of the plurality of filters **412**, a second switch is coupled at a first connection to the second node **422** and at a second connection to a respective filter of the plurality of filters **412**, and a third switch is coupled at a first connection to the third node **424** and at a second connection to a respective filter of the plurality of filters **412**.

[0049] The plurality of filters **412** is coupled to the second plurality of switches **410** and to the third plurality of switches **414**. In some examples, respective filters of the plurality of filters **412** are each coupled to respective switches of the second plurality of switches **410** and to respective switches of the third plurality of switches **414**.

[0050] Respective switches of the third plurality of switches **414** are coupled to either the fourth node **426**, fifth node **428**, sixth node **430**, and/or at least one respective filter of the plurality of filters **412**. In some examples, for the third plurality of switches **414**, a first switch is coupled at a first connection to the fourth node **426** and at a second connection to a respective filter of the plurality of filters **412**, a second switch is coupled at a first connection to the fifth node **428** and at a second connection to a respective filter of the plurality of filters **412**, and a third switch is coupled at a first connection to the sixth node **430** and at a second connection to a respective filter of the plurality of filters **412**.

[0051] The plurality of inputs **401***a* are coupled to the fourth plurality of switches **416**. In some examples, each switch of the fourth plurality of switches **416** is configured to selectively couple to at least one respective input of the plurality of inputs **401***a*, and may be configured to be able to selectively couple to any of the inputs of the plurality of inputs **401***a*. In some examples, a first switch of the fourth plurality of switches **416** is coupled at a first connection to the fourth node **426** and at a second connection to at least one input of the plurality of inputs **401***a*, a second switch of the fourth plurality of switches **416** is coupled at a first connection to the fifth node **428** and at a second connection to at least one input of the plurality of inputs **401***a*, and a third switch of the fourth plurality of switches **416** is coupled at a first connection to the sixth node **430** and at a second connection to at least one input of the plurality of inputs **401***a*.

[0052] The fifth plurality of switches **418** is coupled to one or more inputs of the plurality of inputs **401***a* and is coupled the first node **420**, second node **422**, and third node **424**. The switches of the fifth plurality of switches **418** may be configured to be selectively coupled to any of the first node **420**, second node **422**, or third node **424**. In some examples, the fifth plurality of switches **418** includes a first switch coupled to an input of the plurality of inputs **401***a*, and a second switch coupled to a different input of the plurality of inputs **401***a*. In some examples, the inputs to which the fifth plurality of switches **418** are coupled may be coupled to the output of one or more other modules (for example, the output of a module **302***b*, **304***b*, **306***b* of FIG. **3**).

[0053] The first plurality of switches **408** may be used to provide a bypass path around the second plurality of switches **410**, third plurality of switches **414**, and the plurality of filters **412**. However, a signal arriving at the plurality of inputs **401***a* that routes through the first plurality of switches **408** goes through at least two switches (a switch of the first plurality of switches **408** and a switch of the fourth plurality of switches **416**). By contrast, the fifth plurality of switches **418** can provide a bypass path that includes only one switch (a switch of the fifth plurality of switches **418**).

[0054] The first pair of traces **450** may include a main trace and a coupled trace, wherein the coupled trace is electromagnetically coupled to the main trace (for example, so that a current and/or voltage on the main trace may induce a current and/or voltage on the coupled trace). These traces may be substantially similar to the main trace **106** and coupled trace **108** of FIGS. **1** and **2**. The second pair of traces **452** may include its own respective main trace and coupled trace that are also substantially identical to the main trace **106** and coupled trace **108** of FIGS. **1** and **2**.

[0055] The coupled traces of the pairs of traces **450**, **452** may each have respective "forward" ends and a "reverse" ends corresponding to the mode of operation of a module housing or incorporating the module topology **400**. These respective forward and reverse ends of the coupled traces of the pairs of traces **450**, **452** may be coupled to respective terminals or connections of the first plurality of inputs **401***a*.

[0056] FIG. **5** illustrates a module topology **500** ("module **500**") according to an example. Module **500** is identical to module **400** except that the fifth plurality of switches **418** have been replaced with a plurality of switches **518**. The plurality of switches **518** are coupled to at least one input of the plurality of inputs **401***a*, and are connected to at least one output of the plurality of outputs **401***b*. In some examples, each switch of the plurality of switches **518** may be selectively coupled to any of the outputs of the plurality of outputs **401***b*.

[0057] The plurality of switches **518** therefore provide a bypass path from the plurality of inputs **401***a* to the plurality of outputs **401***b* that avoids the first plurality of switches **408**, second plurality of switches **410**, plurality of filters **412**, third plurality of switches **414**, fourth plurality of switches **416**, and each attenuator **402**, **404**, **406**. That is, the plurality of switches **518** may directly connect one or more inputs of the plurality of inputs **401***a* to one or more outputs of the plurality of outputs **401***b*.

[0058] FIG. **6** illustrates a module topology **600** ("module **600**") according to an example. The module **600** is identical to the module **500** except that the module **600** adds a first impedance network **602**, a second impedance network **604**, and a third impedance network **606**. The impedance networks **602**, **604**, **606** flatten attenuation over frequency, as will be explained with respect to FIG. **7**.

[0059] The first impedance network **602** is coupled to a first output of the plurality of outputs **401***b* and switchably coupled to a reference node (or may be coupled to the reference node and be switchably coupled to the first output of the plurality of outputs **401***b*). The second impedance network **604** is coupled to a second output of the plurality of outputs **401***b* and switchably coupled to a reference node (or may be coupled to the reference node and be switchably coupled to the second output of the plurality of outputs **401***b*). The third impedance network **606** is coupled to a third output of the plurality of outputs **401***b* and switchably coupled to a reference node (or may be coupled to the reference node and be switchably coupled to the third output of the plurality of outputs **401***b*).

[0060] FIG. **7**A illustrates a graph **700** of the transfer functions for various operation modes of a module (for example, a module **200** of FIG. **2**) according to an example. The graph **700** shows frequency on the x-axis and insertion loss on the y-axis. The graph **700** includes a first trace **702** illustrating the transfer function of a module with a bypass switch (such as module **200** of FIG. **2**) in a 0 dB mode compared to a first plurality of traces **704** illustrating the transfer functions of a module without a bypass switch (for example, module **100** of FIG. **1**) in a 0 dB mode and various other modes of operation.

[0061] As can be seen, the slope of the traces of the first plurality of traces **704** decreases faster, on average, than that of the first trace **702**. As a result, as frequency increases, the difference between the first trace **702** and any given trace of the first plurality of traces **704** increases. This is because the bypass route through the bypass switch **202** has different impedance characteristics compared to a route through the source selection switch **114**, filter selection switch **118**, and output selection switch. **122**.

[0062] By comparison, FIG. **7**B illustrates a graph **750** of the transfer functions for various operation modes of a module (for example, a module **200** of FIG. **2**) according to an example. In graph **650**, the first trace **702** and the second plurality of traces **706** all correspond to a module with a bypass switch (such as module **200** of FIG. **2**).

[0063] The addition one or more of the impedance networks **602**, **604**, **606** in FIG. **6** ensures that the "step" between the various traces is constant or relatively constant (e.g., monotonic). For example, in FIG. **7**B, each trace **702**, **706** has approximately the same slope and approximately the same rate of change (e.g., decrease) of the slope, thus ensuring that the difference between the first trace **702** and any trace of the second plurality of traces **706** remains approximately equal over frequency.

[0064] FIG. **8** illustrates a module topology **800** ("module **800**") according to an example. The module **800** is generally identical to module **400** of FIG. **4**, except that switches have been interposed between the attenuators **402**, **404**, **406** and the fifth plurality of switches **418**, and between the attenuators **402**, **404**, **406** and the first node **420**, second node **422**, and/or third node **424**. The switches may reduce off-capacitance of the circuit.

[0065] The module **800** includes a first switch **802**, second switch **804**, third switch **806**, fourth switch **808**, fifth switch **810**, and sixth switch **812**.

[0066] The first switch **802** is coupled between the first node **420** and the first attenuator **402** and is configured to selectively couple the first attenuator **402** to the first node **420**. The second switch **804** is coupled between first node

420 and the fifth plurality of switches **418**, and is configured to selectively couple the fifth plurality of switches **418** to the first node **420**. The third switch **806** is coupled between the second node **422** and the second attenuator **404** and is configured to selectively couple the second attenuator **404** to the second node **422**. The fourth switch **808** is coupled between second node **422** and the fifth plurality of switches **418**, and is configured to selectively couple the fifth plurality of switches **418** to the second node **422**. The fifth switch **810** is coupled between the third node **424** and the third attenuator **406** and is configured to selectively couple the third attenuator **406** to the third node **424**. The sixth switch **812** is coupled between third node **424** and the fifth plurality of switches **418**, and is configured to selectively couple the fifth plurality of switches **418** to the third node **424**.

[0067] FIG. **9** illustrates a module topology **900** ("module **900**") according to an example. The module **900** is generally identical to module **500** of FIG. **5**, except that switches have been interposed between the attenuators **402**, **404**, **406** and the fifth plurality of switches **418**, and between the attenuators **402**, **404**, **406** and plurality of outputs **401***b*. The switches may reduce off-capacitance of the circuit.

[0068] The module **900** includes a first switch **902**, second switch **904**, third switch **906**, fourth switch **908**, fifth switch **910**, and sixth switch **912**.

[0069] The first switch **902** is coupled between the first attenuator **402** and the first output of the plurality of outputs **401***b* and is configured to selectively couple the first attenuator **402** to the first output. The second switch **904** is coupled between the plurality of switches **518** and the first output of the plurality of outputs **401***b* and is configured to selectively couple the plurality of switches **518** to the first output. The third switch **906** is coupled between the second attenuator **404** and the second output of the plurality of outputs **401***b* and is configured to selectively couple the second attenuator **404** to the second output. The fourth switch **908** is coupled between the plurality of switches **518** and the second output of the plurality of outputs **401***b* and is configured to selectively couple the plurality of switches **518** to the second output. The fifth switch **910** is coupled between the third attenuator **406** and the third output of the plurality of outputs **401***b* and is configured to selectively couple the third attenuator **406** to the third output. The sixth switch **912** is coupled between the plurality of switches **518** and the third output of the plurality of outputs **401***b* and is configured to selectively couple the plurality of switches **518** to the third output.

[0070] Examples of the methods and systems discussed herein are not limited in application to the details of construction and the arrangement of components set forth in the following description or illustrated in the accompanying drawings. The methods and systems are capable of implementation in other embodiments and of being practiced or of being carried out in various ways. Examples of specific implementations are provided herein for illustrative purposes only and are not intended to be limiting. In particular, acts, components, elements and features discussed in connection with any one or more examples are not intended to be excluded from a similar role in any other examples.

[0071] Also, the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. Any references to examples, embodiments, components, elements or acts of the systems and methods herein referred to in the singular may also embrace embodiments including a plurality, and any references in plural to any embodiment, component, element or act herein may also embrace embodiments including only a singularity. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or elements. The use herein of "including," "comprising," "having," "containing," "involving," and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

[0072] References to "or" may be construed as inclusive so that any terms described using "or" may indicate any of a single, more than one, and all of the described terms. In addition, in the event of inconsistent usages of terms between this document and documents incorporated herein by reference, the term usage in the incorporated features is supplementary to that of this document; for irreconcilable differences, the term usage in this document controls.

[0073] Various controllers may execute various operations discussed above. Using data stored in associated memory and/or storage, the controller also executes one or more instructions stored on one or more non-transitory computer-readable media, which the controller may include and/or be coupled to, that may result in manipulated data. In some examples, the controller may include one or more processors or other types of controllers. In one example, the controller is or includes at least one processor. In another example, the controller performs at least a portion of the operations discussed above using an application-specific integrated circuit tailored to perform particular operations in addition to, or in lieu of, a general-purpose processor. As illustrated by these examples, examples in accordance with the present disclosure may perform the operations described herein using many specific combinations of hardware and software and the disclosure is not limited to any particular combination of hardware and software components. Examples of the disclosure may include a computer-program product configured to execute methods, processes, and/or operations discussed above. The computer-program product may be, or include, one or more controllers and/or processors configured to execute instructions to perform methods, processes, and/or operations discussed above.

[0074] Having thus described several aspects of at least one embodiment, it is to be appreciated various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to be part of, and within the spirit and scope of, this disclosure. Accordingly, the foregoing description and drawings are by way of example only.

What is claimed is:

1. A module for an electronic telecommunications device, comprising:

a forward input;

a reverse input;

a bypass input;

an output;

a first plurality of switches coupled between the forward input, the reverse input, the bypass input, and the output, the first plurality of switches configured to selectively couple at least one of the forward input, reverse input, or bypass input to the output; and

at least one bypass switch configured to selectively couple the bypass input to the output.

2. The module of claim **1** wherein the first plurality of switches includes a first switch, a second switch, and a third switch, the first switch being coupled to the forward input, reverse input, and bypass input, the second switch being coupled to the first switch and the third switch, and the third switch being coupled between the second switch and the output.

3. The module of claim **2** further comprising a plurality of filters coupled to the second switch and the third switch, wherein the second switch is configured to selectively couple the first switch to a filter of the plurality of filters or to the third switch, and the third switch is configured to selectively couple the filter of the plurality of filters or the second switch to the output.

4. The module of claim **1** further comprising an attenuator coupled between the plurality of switches and the output, wherein the at least one bypass switch is coupled either between the plurality of switches and the attenuator, or between the attenuator and the output.

5. The module of claim **4** further comprising:

one or more additional outputs and one or more additional attenuators, the one or more additional attenuators being coupled between the plurality of switches and the one or more additional outputs; and

one or more additional bypass inputs coupled to the at least one bypass switch, the at least one bypass switch being configured to selectively couple at least one of the one or more additional bypass inputs to the output.

6. The module of claim **5** wherein the at least one bypass switch is further coupled either between the one or more additional attenuators and the plurality of switches or between the one or more additional attenuators and the one or more additional outputs.

7. The module of claim **5** further comprising a plurality of impedance networks, each impedance network coupled to a respective attenuator or additional attenuator and to a respective output or additional output.

8. The module of claim **4** further comprising an impedance network coupled between the attenuator and the output.

9. The module of claim **1** further comprising a second plurality of switches coupled to the output, the second plurality of switches including at least one switch coupled between the output and the at least one bypass switch and at least one switch coupled between the output and the plurality of switches.

10. The module of claim **9** further comprising an attenuator coupled either between the second plurality of switches and the output or between the second plurality of switches and the plurality of switches.

11. The module of claim **1** wherein the plurality of switches includes a first switch, a second switch, a third switch, and a fourth switch, the first switch coupled to the second switch, the second switch coupled to the third switch, the third switch coupled to the output and the first switch coupled to at least one of the forward input, reverse input, or bypass input, and the fourth switch coupled to the first switch and to the output.

12. The module of claim **1** further comprising:

a first trace configured to receive a signal;

a second trace electromagnetically coupled to the first trace and configured to provide a coupled signal responsive to the first trace receiving the signal, wherein the second trace is coupled at a first end to the forward input and at a second end to the reverse input.

13. The module of claim **12** further comprising:

at least on additional first trace configured to receive a signal; and

at least one additional second trace electromagnetically coupled to the at least one first trace and configured to provide at least one additional coupled signal respective to the at least one first trace receiving at least one additional signal, wherein the at least one second trace is coupled at a first end to at least one additional forward input and at a second end to at least one additional reverse input.

14. A system of connected modules comprising:

a first module including a first forward input, a first reverse input, a first bypass input, a first output, a first plurality of switches coupled between the first forward input, first reverse input, and first output, and at least one first bypass switch coupled between the first bypass input and the first output; and

a second module including a second forward input, a second reverse input, a second bypass input coupled to the first output, a second output, a second plurality of switches coupled between the second forward input, second reverse input, and second output, and at least one second bypass switch coupled between the second bypass input and the second output.

15. The system of claim **14** wherein the first module includes a first impedance network coupled to the first output and the second module includes a second impedance network coupled to the second output.

16. The system of claim **15** wherein the first module and second module are configured to operate in a plurality of modes, including a first mode, a second mode, and a third mode, and wherein a first impedance of the first impedance network and a second impedance of the second impedance network are chosen such that a first transfer function of the system in the first mode changes at approximately a same rate as both a second transfer function of the system in the second mode and a third transfer function of the system in the third mode.

17. The system of claim **14** wherein the first module includes a first attenuator coupled to the first output, and the second module includes a second attenuator coupled to the second output.

18. The system of claim **14** wherein the first module includes a first plurality of filters coupled to the first plurality of switches, and the second module includes a second plurality of filters coupled to the second plurality of switches.

19. The system of claim **18** wherein the first module includes a first auxiliary bypass switch coupled to the first plurality of switches and configured to selectively bypass the first plurality of filters and to selectively couple at least one of the first forward input or first reverse input to the first output.

20. A system of connected modules comprising:

a plurality of modules coupled together in a daisy chain, each respective module of the plurality of modules having a forward input, a reverse input, a bypass input, an output, a plurality of switches coupled between the forward input, the reverse input, and the output of the respective module, and at least one bypass switch coupled between the bypass input and the output of the respective module, the output of a first module of the plurality of modules in the series being coupled to the bypass input of a second module of the plurality of modules in the series.

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